1- CLASS TASKS SOLUTIONS

--- TASK 1, 2, 3 & 4 SOLUTION KEY (PYTHON CODE) ---

```Python
import pandas as pd

# Task 1: Setup & Structural Inspection
raw_orders = {
    "order_id": ["ORD-201", "ORD-202", "ORD-203", "ORD-204", "ORD-205", "ORD-206", "ORD-207", "ORD-208"],
    "date": ["2026-10-10", "2026-10-10", "2026-10-11", "2026-10-11", "2026-10-12", "2026-10-12", "2026-10-13", "2026-10-13"],
    "category": ["Electronics", "Apparel", "Home", "Electronics", "Apparel", "Home", "Electronics", "Apparel"],
    "quantity": [1, 3, 2, 1, 5, 4, 2, 1],
    "revenue": [1200.00, 150.00, 80.00, 899.00, 250.00, 160.00, 1798.00, 75.00],
    "payment_method": ["Credit Card", "PayPal", "Credit Card", "Debit Card", "PayPal", "Credit Card", "Credit Card", "Debit Card"]
}

df_orders = pd.DataFrame(raw_orders)

print("Shape (Rows, Cols):", df_orders.shape) # (8, 6)
print("\nFirst 4 Rows:\n", df_orders.head(4))
print("\nLast 2 Rows:\n", df_orders.tail(2))

# Task 2: Metadata & Column Selection
print("\n--- INFO ---")
df_orders.info()
print("\nColumns:", df_orders.columns)

df_revenue = df_orders[["order_id", "category", "revenue"]]
print("\nSub-DataFrame:\n", df_revenue)

# Task 3: Summary Statistics
print("\n--- DESCRIBE ---")
print(df_orders.describe())

total_rev = df_orders["revenue"].sum()
mean_rev = df_orders["revenue"].mean()
min_rev = df_orders["revenue"].min()
max_rev = df_orders["revenue"].max()

print(f"\nTotal Revenue: ${total_rev:.2f}")  # $4612.00
print(f"Mean Revenue:  ${mean_rev:.2f}")   # $576.50
print(f"Min Revenue:   ${min_rev:.2f}")    # $80.00
print(f"Max Revenue:   ${max_rev:.2f}")    # $1798.00
```

--- TASK 4 BUSINESS SUMMARY SOLUTION KEY ---

"To: Samantha Reed, E-Commerce Analytics Manager Re: E-Commerce Transaction Dataset Discovery

The batch transaction export was loaded and audited successfully. The dataset contains 8 orders across 6 attributes, generating a total revenue of $4,612.00 with an average order value of $576.50. Metadata auditing confirms all 8 rows contain non-null values; however, the `date` column is currently stored as a string (`object`) rather than datetime format. The dataset is structurally sound and ready for downstream analysis."

2- HOMEWORK SOLUTION

```python
# Homework Problem 1 & 2 Python Solution
import pandas as pd

shipment_data = {
    "shipment_id": ["SHP-801", "SHP-802", "SHP-803", "SHP-804", "SHP-805",
"SHP-806", "SHP-807", "SHP-808", "SHP-809", "SHP-810"],
    "vendor_name": ["ApexLogistics", "GlobalFreight", "ApexLogistics",
"FastShip", "GlobalFreight", "FastShip", "ApexLogistics", "GlobalFreight",
"FastShip", "ApexLogistics"],
    "destination_dc": ["DC-Alpha", "DC-Beta", "DC-Alpha", "DC-Gamma", "DC-
Beta", "DC-Gamma", "DC-Alpha", "DC-Beta", "DC-Gamma", "DC-Alpha"],
    "weight_lbs": [1250, 3400, 2100, 850, 4200, 1100, 3100, 2900, 950,
2750],
    "freight_cost": [450.00, 1120.00, 720.00, 310.00, 1450.00, 400.00,
1050.00, 980.00, 340.00, 910.00],
    "delivery_status": ["Delivered", "Delivered", "In Transit",
"Delivered", "Delayed", "Delivered", "In Transit", "Delivered",
"Delivered", "In Transit"]
}

df_shipments = pd.DataFrame(shipment_data)

shape = df_shipments.shape # (10, 6)
total_freight = df_shipments["freight_cost"].sum() # 7730.00
avg_weight = df_shipments["weight_lbs"].mean()    # 2260.00
max_freight = df_shipments["freight_cost"].max()   # 1450.00

print("Shape:", shape)
print("Total Freight Cost: $", total_freight)
print("Average Weight (lbs):", avg_weight)
print("Max Freight Cost: $", max_freight)
```

- **Homework Grading Criteria:**
  - Correctly identifies dataset dimensions as 10 rows by 6 columns.
  - Calculates total freight cost = $7,730.00, average weight = 2,260 lbs, max freight cost = $1,450.00.

- Confirms metadata health: all 10 records are complete with no missing values, `weight_lbs` as `int64`, `freight_cost` as `float64`, and text fields as `object`.